# Anatomically-Constrained LDMs for Rare Periodontitis Case Augmentation

Rafal Ciolek - 2248504
Computer Vision
Sapienza University of Rome

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# Class Imbalance Meets Anatomical Hallucination

**Severe cases are rare.** In public periodontitis datasets, Stage IV (severe) bone loss is dramatically underrepresented compared to Stage I - so models see far fewer examples of exactly the cases that matter most clinically.

**Generative models hallucinate anatomy.** Standard DDPMs/LDMs can synthesize realistic-looking X-rays whose tooth and bone structures are geometrically implausible - dangerous in a clinical augmentation pipeline.

**Our goal:** a Latent Diffusion Model whose rare-case generation is strictly constrained by anatomical segmentation masks - so synthetic data is safe to add to a real training set.

# Dataset

**Periodontal Bone Loss Keypoint Dataset (perio_KPT)**

**163 / 29 / 15**

train / val / held-out external test images

**Annotations:** bounding boxes, 11 keypoints/tooth, instance segmentation masks (7 classes), and a derived PBL (percent bone loss) map per tooth.

**Rare conditions tracked:** ARR (apical root resorption), PLS (periodontal ligament space widening), high PBL (≥0.33), and furcation involvement.

**Split:** stratified by severity bin so the rarest Stage-IV cases appear in both train and val, not left to chance.

# Baseline Pipeline: Unconstrained LDM

### Stage 1: VAE

256x256 to 32x32x4 latent (f=8), grayscale, from scratch

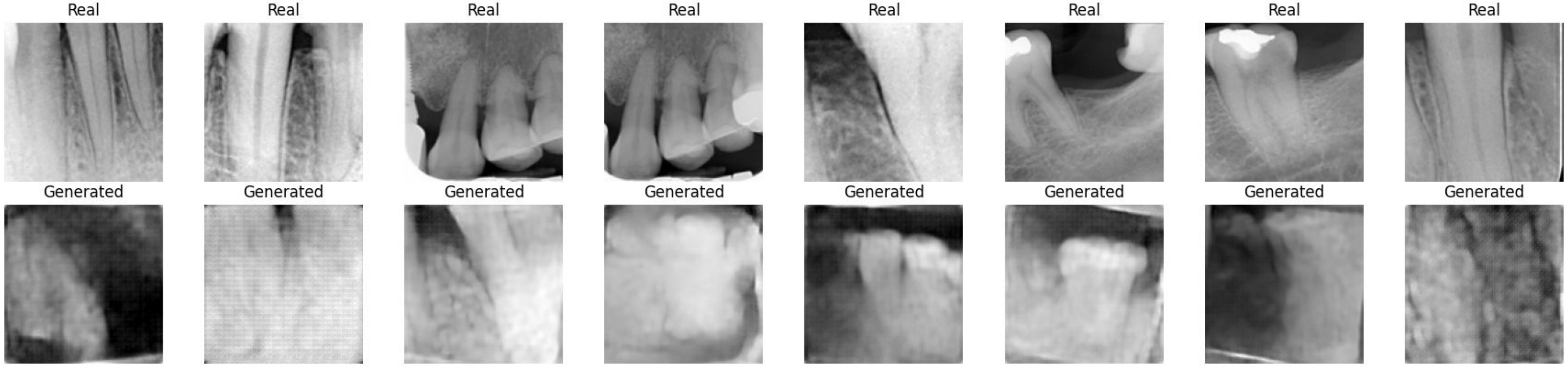
### Stage 3: Sample

50-step DDIM sampling, decoded via the frozen VAE into a rare-lesion

### Stage 2: Diffusion

1000-step latent U-Net, cosine schedule, EMA weights - 200 epochs on rare-patch latents

# Baseline Results: Anatomical Hallucination




**Structural error analysis - generated vs. real (unconstrained):**

Edge density 0.29 vs 0.47 • Gradient energy 0.016 vs 0.058 • SSIM-to-nearest-real 0.59 • Bone-band contrast 0.025 vs 0.031

*Net effect: boundary blur, texture inconsistency, global structure mismatch, bone-level ambiguity*

# Novelty: Anatomical Guidance

## Full-resolution mask conditioning via zero-convs

The baseline's structural errors trace back to one cause: mask signal squeezed through a 32x32 latent bottleneck loses exactly the fine boundaries that matter clinically.

**Own conditioning tower.** Mask + PBL stay at full 256x256 resolution, encoded through a dedicated tower matching every U-Net resolution.

**Zero-conv injection.** Each stage injects additively into the matching U-Net decoder stage via a zero-initialized conv (ControlNet) - a clean identity at step 0 for stable training.

**Also fixed:** NEAREST resizing for masks/PBL, so class labels never blend across anatomical edges.

# Data Flow: From Raw Inputs to Generated Output

**IMAGE PATH**

Real X-ray image → to 256x256x1 → VAE encoder to 32x32x4 latent

**MASK/PBL PATH**

Mask + PBL map → to 256x256 (NEAREST) → One-hot (7ch) + PBL into Condition Encoder

**Latent U-Net: latent + conditioning, 25-step DDIM denoising** → Denoised latent $z_0$ (32x32x4) → VAE decoder to 256x256 generated image

*The VAE encoder (image path) and VAE decoder (final step) are the same frozen VAE, trained once in Stage 1 and reused for every diffusion step during sampling. Both paths are letterboxed to identical 256×256 resolution before entering the network.*

# Architecture: Mask-Guided Injection

### Mask/PBL Encoder

256x256 mask+PBL into its own encoder, downsampled to 4x4

### Constrained Sample

Guided latent, decoded via VAE into a constrained image

### Zero-Conv Injection

Added additively into the matching-resolution U-Net decoder stage via a zero-init conv

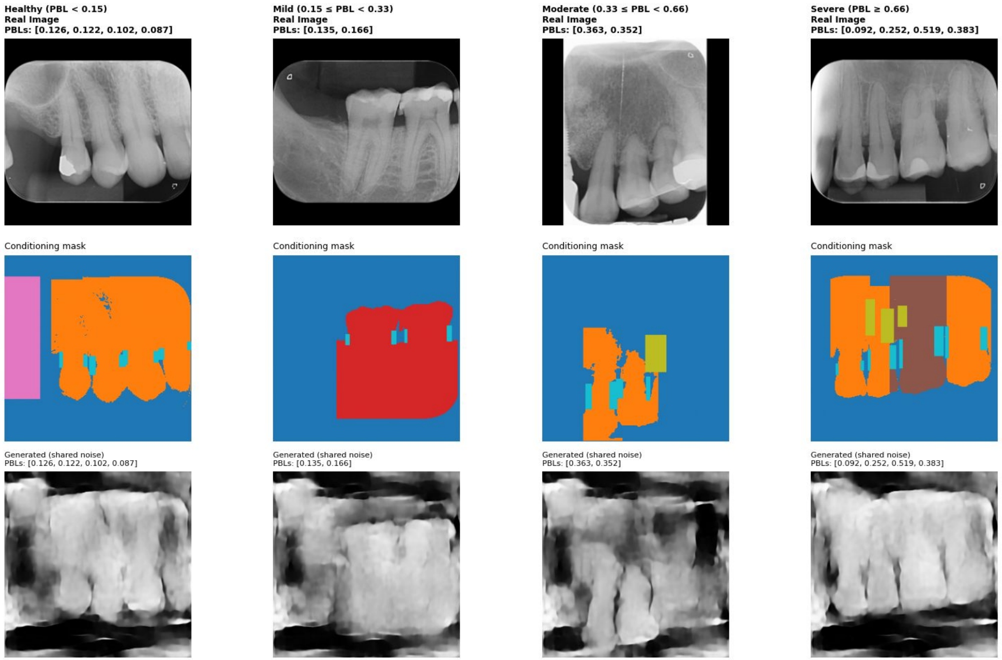
# Training Validation Scores

| Component | Baseline (unconstrained) | Mask-Guided (constrained) |
|---|---|---|
| **VAE - best val L1 reconstruction loss** | 0.0235 | 0.0169 |
| **Diffusion - best val noise-prediction MSE** | 0.134 | 0.077 |

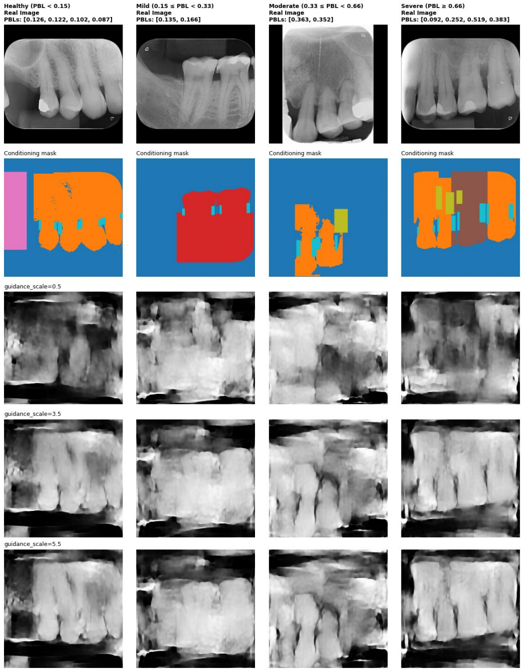
*Lower is better for both metrics. Baseline trains on cropped rare patches only (128px); mask-guided trains on the full periapical dataset (256px) with mask/PBL conditioning.*

**Both components converge to a lower validation loss under the constrained pipeline - and the diffusion model needs only 50 epochs (vs 200) to get there.**

# Constrained Results Across Severity Classes



# Ablation: Guidance Scale Trade-off




Rows (top to bottom): real image, mask, then classifier-free guidance at scale 0.5, 3.5, and 5.5 - same seed and DDIM steps throughout, so guidance strength is the only variable.

**scale = 0.5:** weak conditioning, mask is mostly ignored.

**scale = 3.5 (chosen):** best balance - sharp, mask-faithful boundaries without losing diversity.

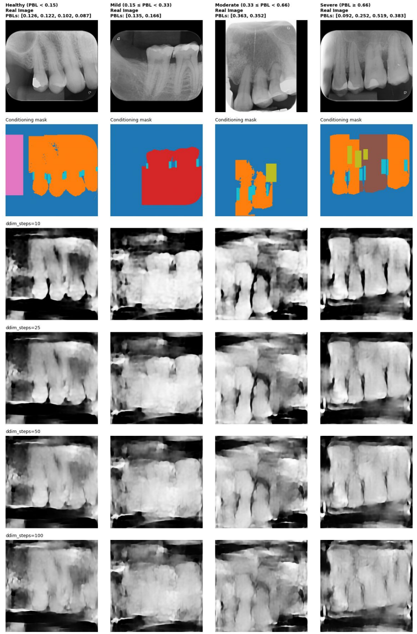
**scale = 5.5:** over-constrained - saturated contrast, loss of fine texture diversity.

# Ablation: Sampling Steps

## 25

DDIM steps chosen for generation - half the compute of the 50-step default.

Tested 10 / 25 / 50 / 100 steps at fixed guidance_scale, same seed. Quality plateaus well before 100 - 25 steps is visually indistinguishable from 50.



# Downstream Impact: YOLOv8-Pose Detection

163 real train images vs. 163 real + 214 synthetic (107 rare images × 2 variants) - evaluated on an identical, untouched 15-image external held-out test set.

| Class | mAP50 real-only | mAP50 real+synth | Δ |
|---|---|---|---|
| Single Root | 0.899 | 0.810 | **−0.089** |
| Double Root | 0.848 | 0.963 | **+0.115** |
| Triple Root | 0.830 | 0.597 | **−0.233** |
| PLS | 0.134 | 0.165 | **+0.031** |
| **ARR (rarest)** | 0.084 | 0.369 | **+0.285** |
| **Overall** | 0.559 | 0.581 | **+0.022** |

ARR - the rarest lesion class - jumps 4.4x in mAP50 (0.084 to 0.369). Overall mAP50 improves 0.559 to 0.581 and mAP50-95 improves 0.194 to 0.216.

# Discussion & Limitations

**Not every class benefits equally.** Single Root and Triple Root regress (-0.089, -0.233 mAP50) - synthetic data is heavily skewed toward ARR/PLS tags, so well-represented classes see mostly noise from augmentation, not signal.

**Tiny rare-case counts.** Only ~15 true Stage-IV images exist in the whole dataset; the diffusion latent bank leans on aggressive patch augmentation (x16) to compensate - more real severe cases would help more than any architectural change.

**Next steps:** class-balanced synthetic sampling (not just tag-driven), per-class guidance-scale tuning, and validation on a larger external, multi-clinic test set. Building bigger model, the current one use not more than GB of VRAM During Training

# References

- **Papers:**

  - Ho, Jain & Abbeel. Denoising Diffusion Probabilistic Models. NeurIPS 2020.
  - Kazerouni et al. Diffusion Models in Medical Imaging: A Comprehensive Survey. Medical Image Analysis, 2023.
  - Pinaya et al. Brain Imaging Generation with Latent Diffusion Models. MICCAI Workshop DGM4MICCAI, 2022.
  - Konz, Chen, Dong & Mazurowski. Anatomically-Controllable Medical Image Generation with Segmentation-Guided Diffusion Models. MICCAI 2024.

# Thank you for the attention!